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Improving Learning from Visual Demonstration Methods by Target Localization

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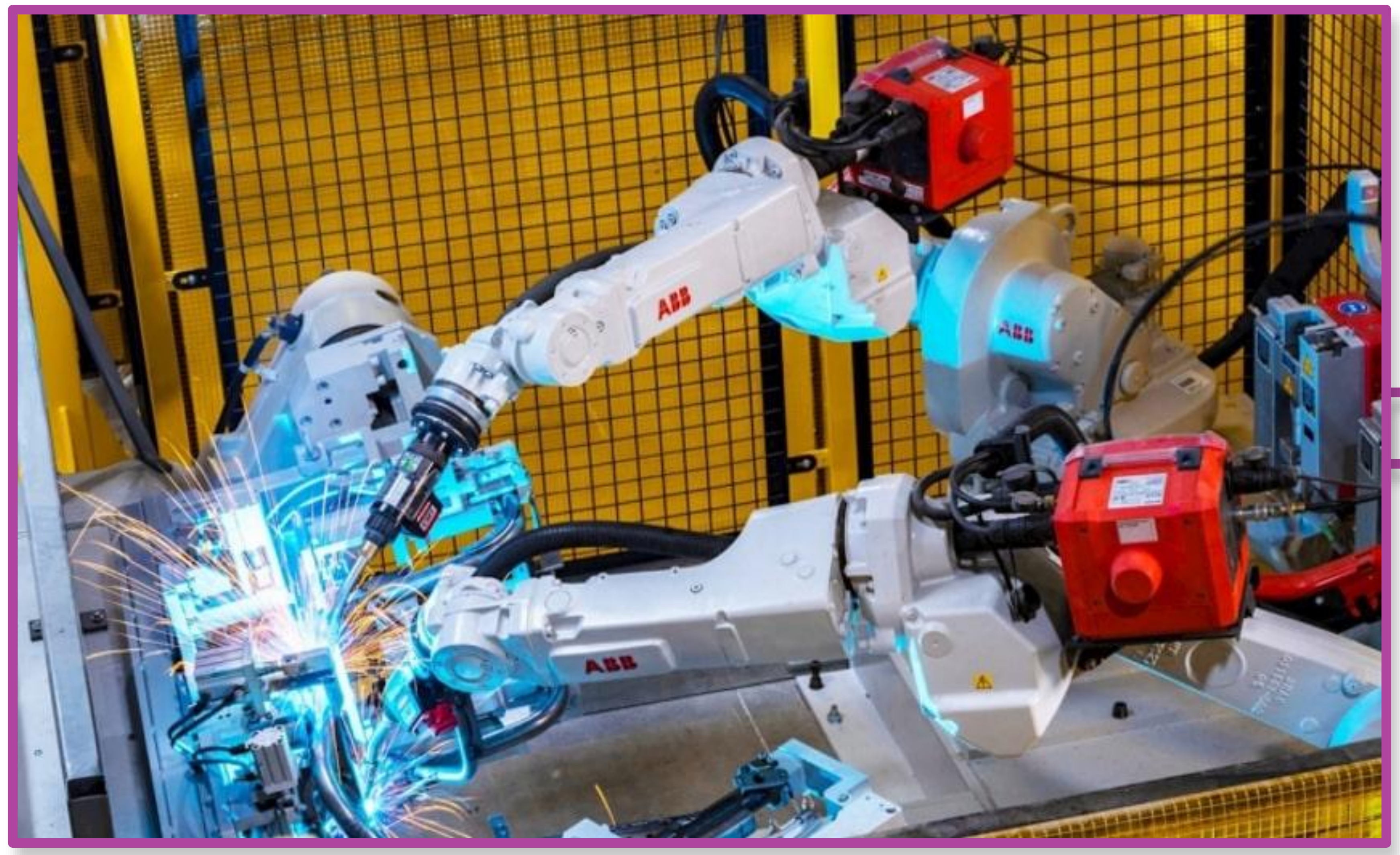
Context

- Robots are asked to work in **novel scenarios**:
  - They are not requested to perform fixed and repetitive operations over time, but to solve **multiple prompted** tasks *[Brohan et al.]*

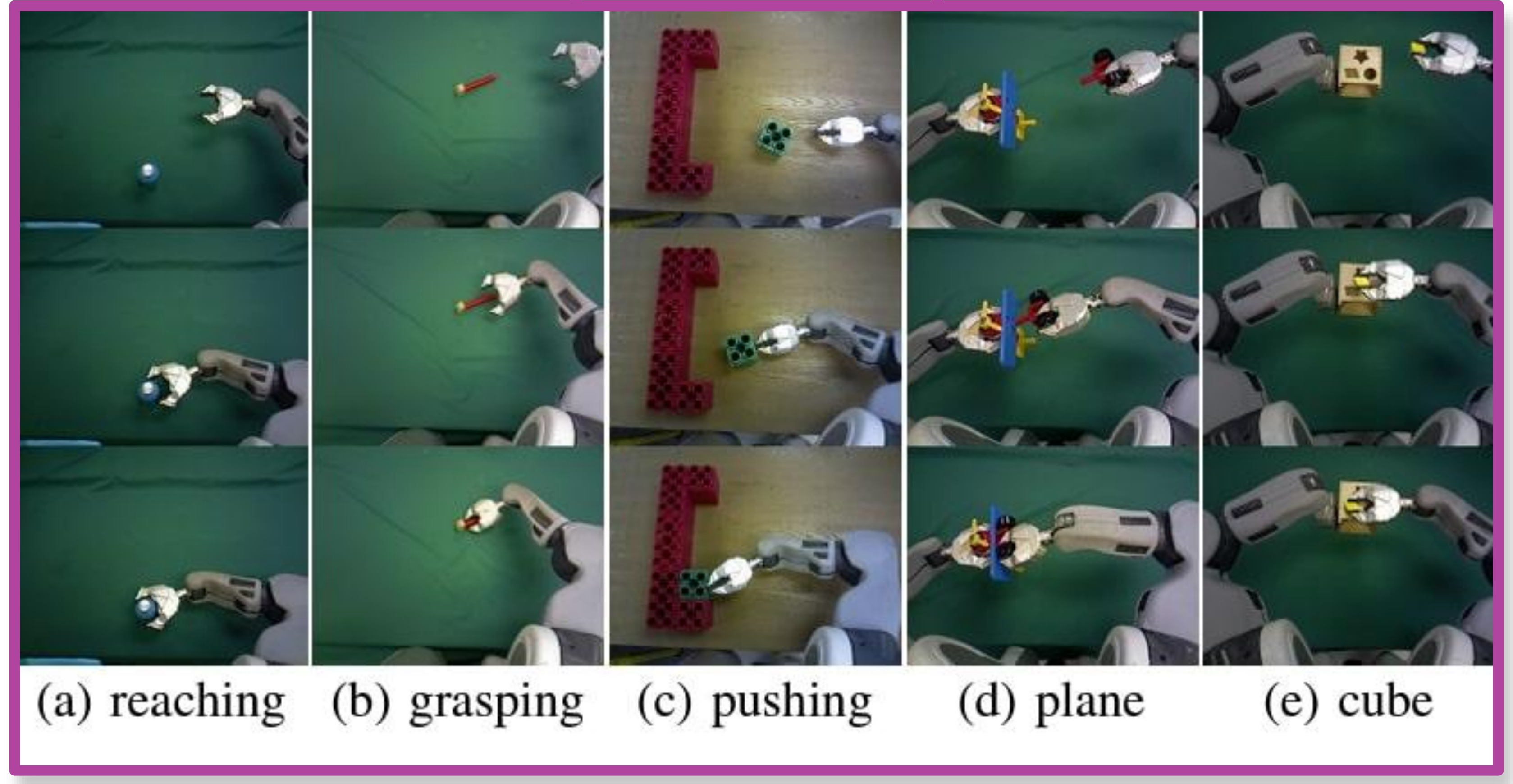


Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., … Zitkovich, B. (2023, July). RT-1: Robotics Transformer for Real-World Control at Scale. Proceedings of Robotics: Science and Systems. doi:10.15607/RSS.2023.XIX.025

Context

- We want to move from a setting where we train a model for each task *[Zhang et al]* to a setting where we train a single model able to solve multiple tasks *[Jang et al]*
- The way how we **inform** the control policy with the task to be performed is crucial

Zhang et al

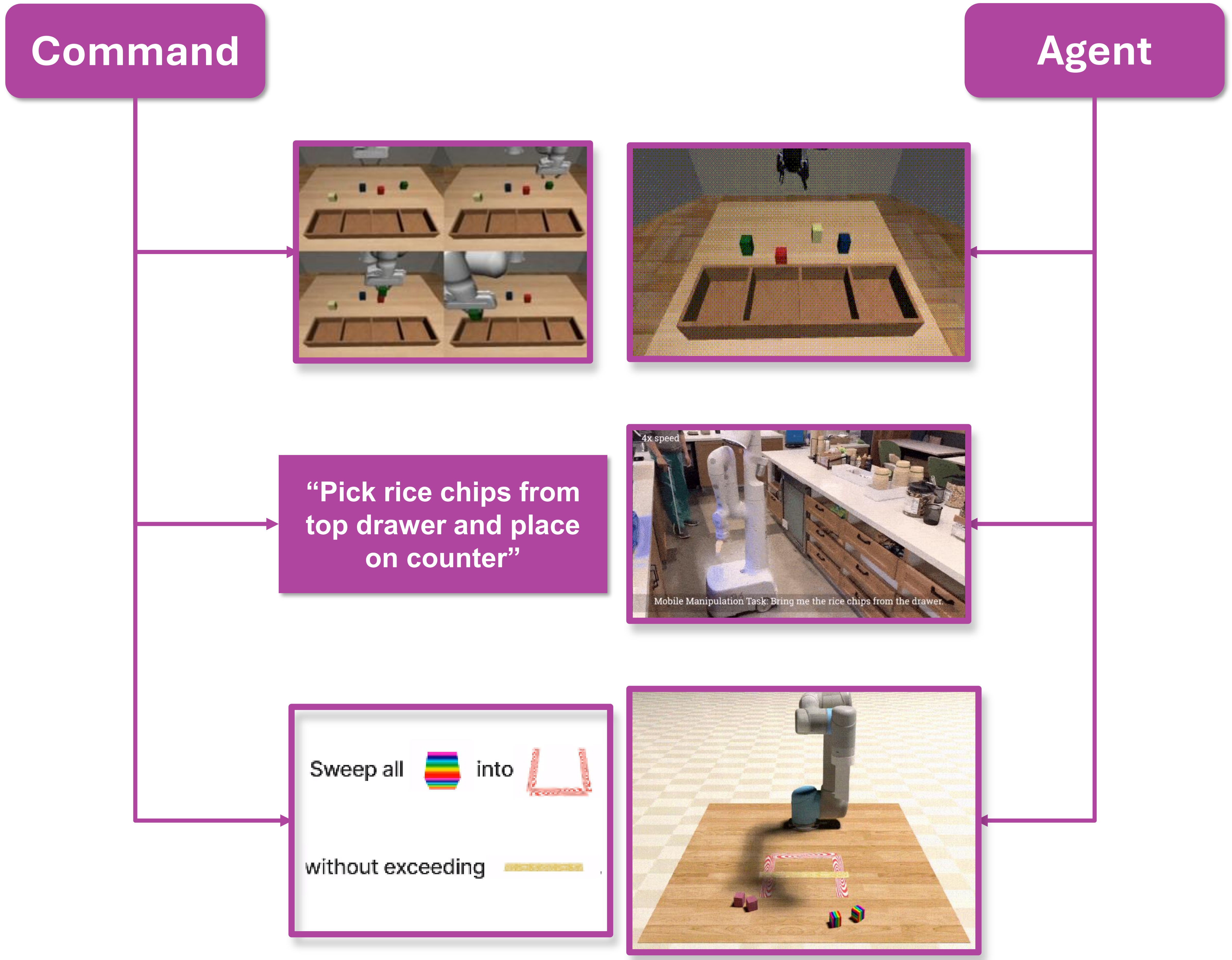


Jang et al

Zhang, T., McCarthy, Z., Jow, O., Lee, D., Chen, X., Goldberg, K., & Abbeel, P. (2018, May). Deep imitation learning for complex manipulation tasks from virtual reality teleoperation. In 2018 IEEE international conference on robotics and automation (ICRA) (pp. 5628-5635). IEEE.
Jang, E., Irpan, A., Khansari, M., Kappler, D., Ebert, F., Lynch, C., ... & Finn, C. (2022, January). Bc-z: Zero-shot task generalization with robotic imitation learning. In Conference on Robot Learning (pp. 991-1002). PMLR.

Context:
Multi-Task Imitation Learning

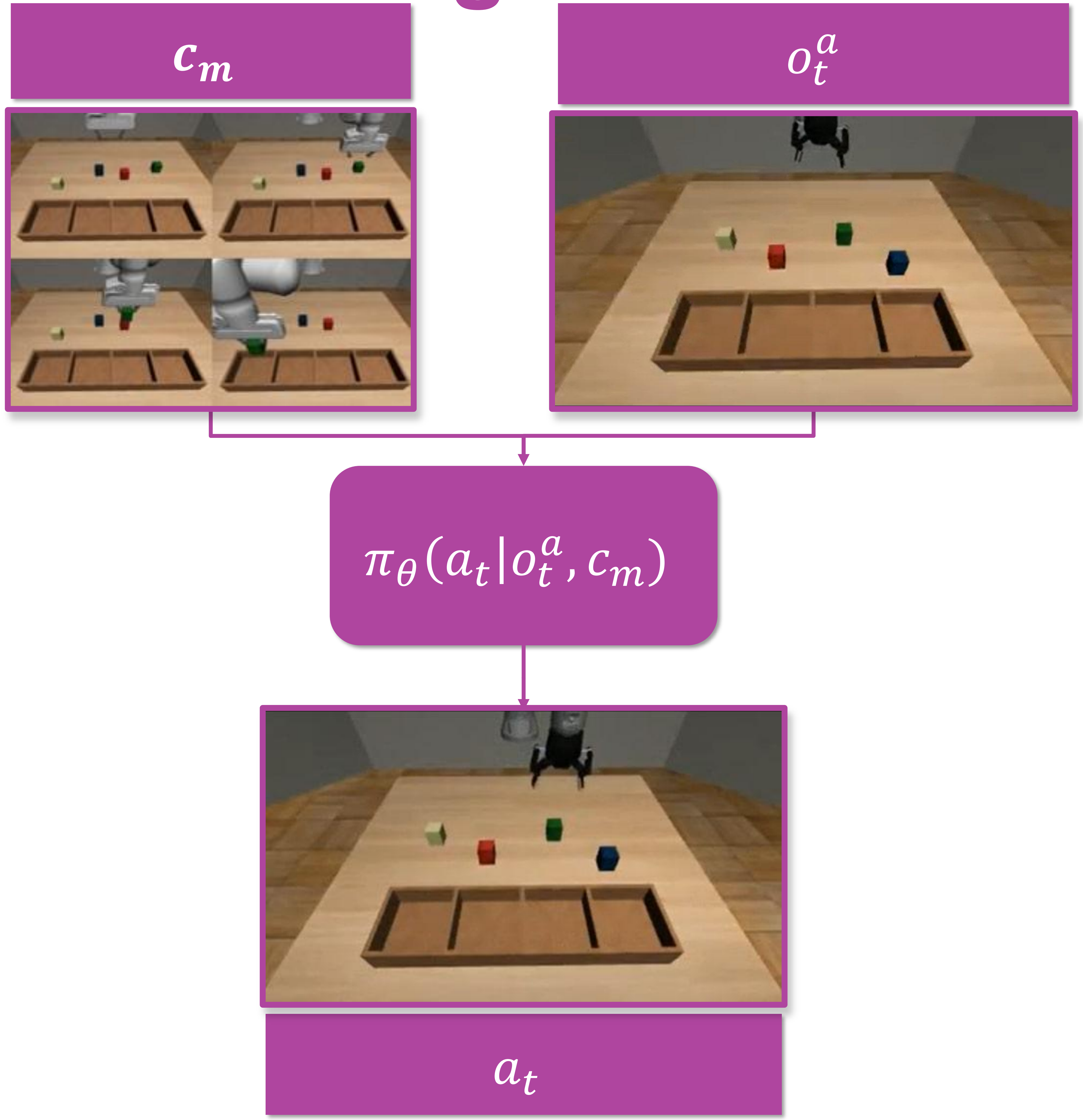
- The command can be defined as:
  - ***Expert video-demonstrations*** *[Mandi et al]*, use video of task execution to describe it
  - *Natural Language description [Brohan at al]*, use text to describe the desired task
  - *Multi-Modal prompt [Yunfan et al]*, use text and visual information to describe the task



Mandi, Z., Liu, F., Lee, K., & Abbeel, P. (2022, May). Towards more generalizable one-shot visual imitation learning. In 2022 International Conference on Robotics and Automation (ICRA) (pp. 2434-2444). IEEE.
Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., … Zitkovich, B. (2023, July). RT-1: Robotics Transformer for Real-World Control at Scale. Proceedings of Robotics: Science and Systems. doi:10.15607/RSS.2023.XIX.025
Yunfan Jiang, Agrim Gupta, Zichen Zhang, Guanzhi Wang, Yongqiang Dou, Yanjun Chen, Li Fei-Fei, Anima Anandkumar, Yuke Zhu, & Linxi Fan (2023). VIMA: General Robot Manipulation with Multimodal Prompts. In Fortieth International Conference on Machine Learning

Context:
Multi-Task Imitation Learning

- How to easily program a robot to perform requested tasks?
  - Learn a **parametrized control-policy** π_θ, given in input a demonstration video c_m and the agent visual state o_t^a, generates the corresponding action a_t
  - *Multi-Task Imitation-Learning* has proven to be an effective approach to learn π_θ



Context:
Multi-Task Imitation Learning

- There is a set of n **distinct tasks** $T = \{T_1, T_2, \dots, T_n\}$, where the $i-th$ task has M_i **task variations**
- For each task, we build a **demonstration dataset** $D_i = ((c_m, \tau_m), m \in M_i)$
  - **Demonstrator video**:

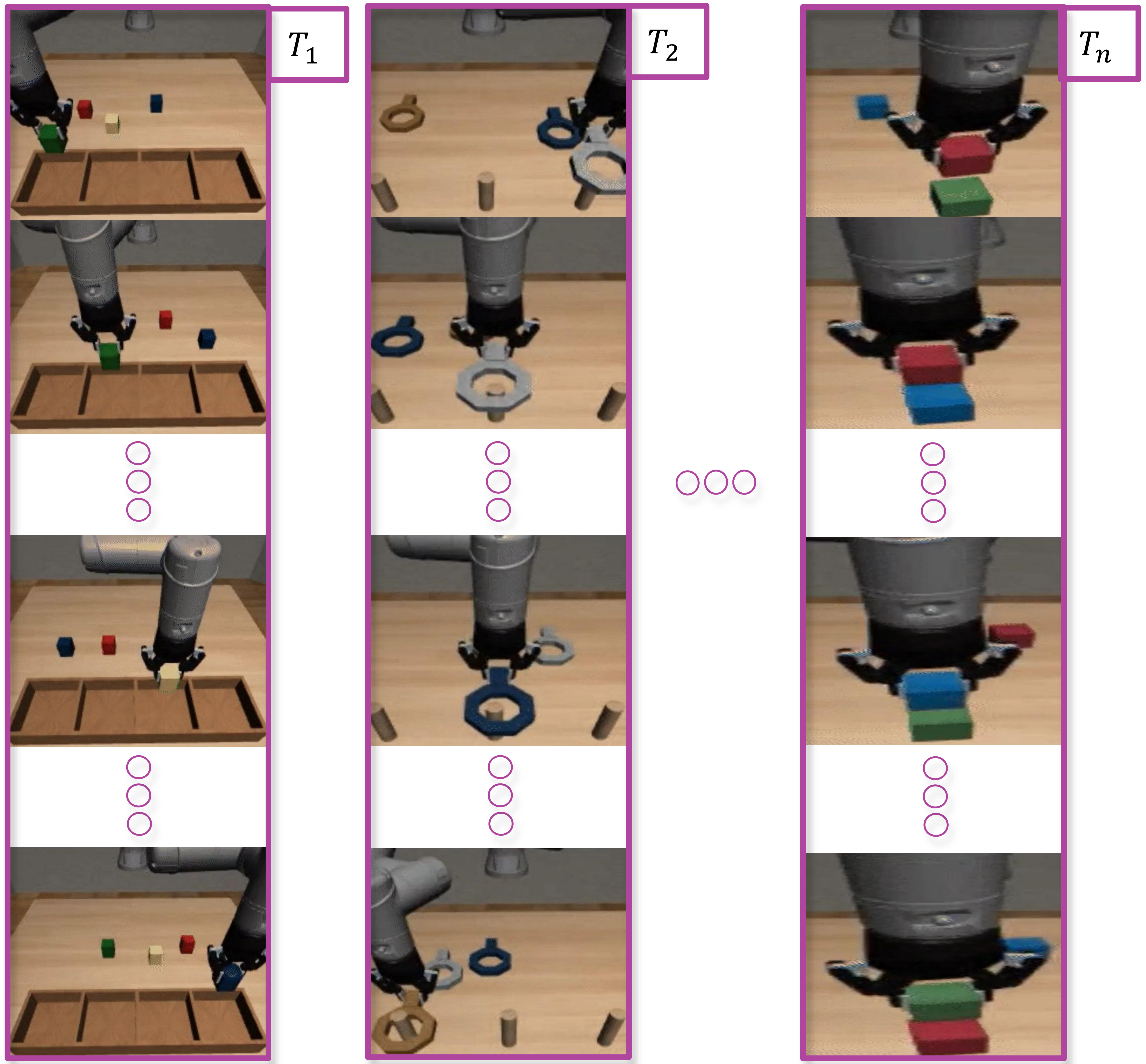
$$c_m = \left\{o_1^d, o_2^d, \dots, o^d{}_{T'}\right\}$$

  - **Agent Trajectories**:

$$\tau_m = \{(o_1^a, a_1), (o_2^a, a_2), \dots, (o_T^a, a_T)\}$$

- **Supervised-Learning Procedure:**

$$\theta^* = \underset{\theta}{\operatorname{argmin}}\, L(\pi_\theta(a_t | o_t^a, c_m))$$



Context:
SoTA Evaluation

- We evaluate two SoTA methods for the Video-Conditioned Multi-Task Imitation Learning TOSIL *[Dasari et al]* and MOSAIC *[Mandi et al]*

- These methods can perform **valid trajectory** but manipulating **wrong objects**

Task	Method	Reaching [%]	Success [%]	Wrong Obj. [%]
Pick-Place	TOSIL	33.12±1.08	26.87±0.62	66.04±1.57
	MOSAIC	62.90±0.95	58.75±1.87	37.71±0.72
Nut-Assembly	TOSIL	36.30±1.28	31.11±1.92	56.30±3.57
	MOSAIC	38.89±1.11	33.33±1.11	51.48±2.31
Stack-Block	TOSIL	69.44±0.96	48.89±2.54	41.67±1.66
	MOSAIC	60.56±0.96	53.33±0.16	36.11±0.96
Button	TOSIL	83.33±1.66	75.00±3.33	16.67±1.67
	MOSAIC	100.00±0.00	100.00±0.00	0.00±0.00

Dasari, S., & Gupta, A. (2021, October). Transformers for one-shot visual imitation. In Conference on Robot Learning (pp. 2071-2084). PMLR.Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., … Zitkovich, B. (2023, July). RT-1: Robotics Transformer for Real-World Control at Scale. Proceedings of Robotics: Science and Systems. doi:10.15607/RSS.2023.XIX.025

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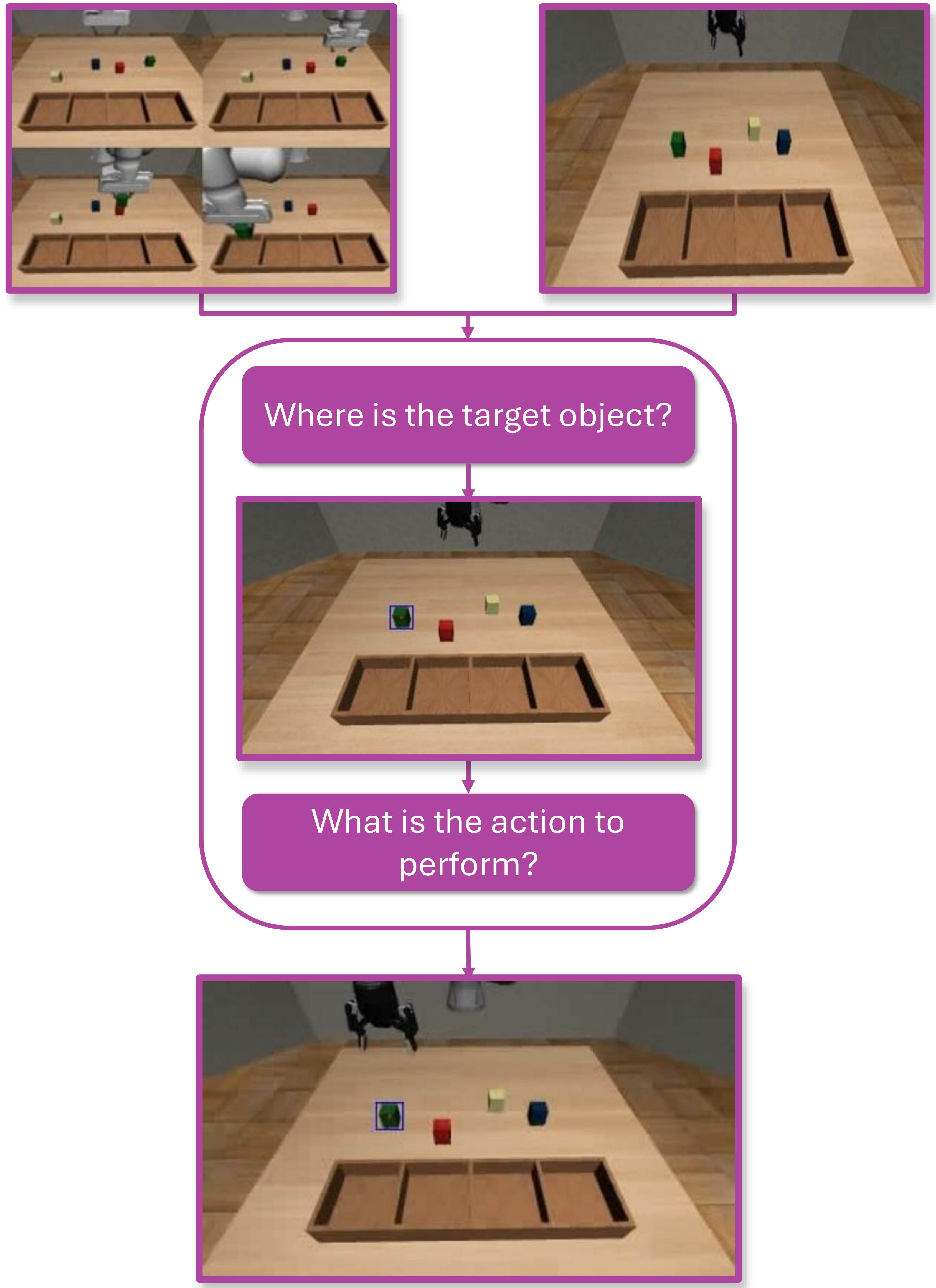
Context:
SoTA Evaluation

- The manipulation decision-making problem is composed of two sub-problems: **Command Analysis** and **Action Generation**
- *Command Analysis:* **cognitive task**, where the system must identify the demonstrated intent and the object of interest
- *Action Generation:* **control task**, the system must generate valid control actions according to the current agent state and prompted command
- SoTA methods propose **end-to-end architectures**



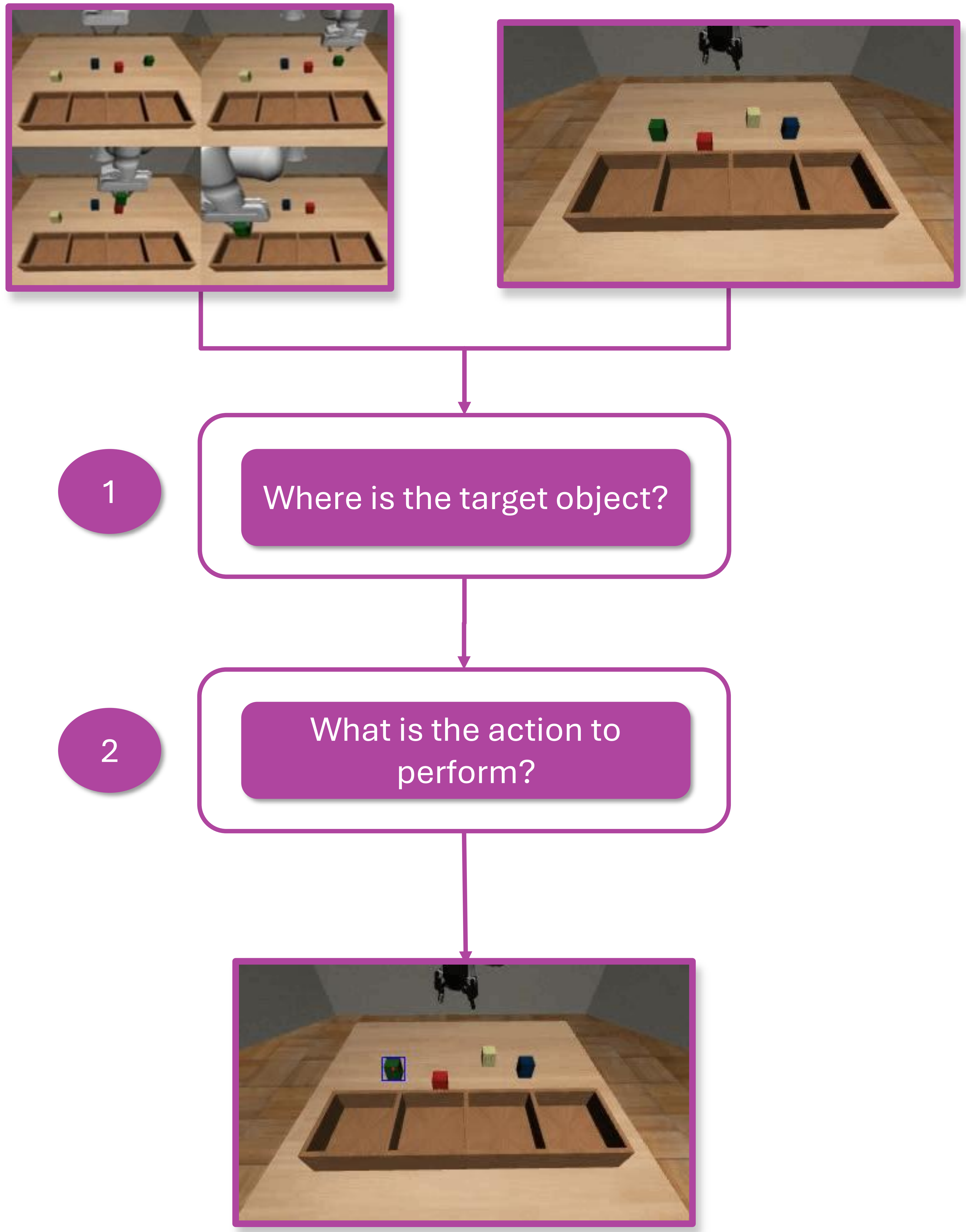
Context:
SoTA Evaluation

- The end-to-end architectures are trained according to the ***action-centric behavioral cloning loss***
- Despite this **misalignment** between the **optimization goal** and the **desired behavior**, in SoTA methods is assumed that the model **learns to encode** both the task intent and the current state of the environment (e.g., position of target object)

Task	Method	Reaching [%]	Success [%]	Wrong Obj. [%]
Pick-Place	TOSIL	33.12±1.08	26.87±0.62	66.04±1.57
	MOSAIC	62.90±0.95	58.75±1.87	37.71±0.72
	MOSAIC (2 GT)	89.17±1.30	**84.17±1.57**	11.46±1.30
Nut-Assembly	TOSIL	36.30±1.28	31.11±1.92	56.30±3.57
	MOSAIC	38.89±1.11	33.33±1.11	51.48±2.31
	MOSAIC (2 GT)	85.19±4.49	**78.89±4.01**	11.11±2.22
Stack-Block	TOSIL	69.44±0.96	48.89±2.54	41.67±1.66
	MOSAIC	60.56±0.96	53.33±0.16	36.11±0.96
	MOSAIC (2 GT)	91.11±0.96	**73.89±4.19**	7.22±0.96
Button	TOSIL	83.33±1.66	75.00±3.33	16.67±1.67
	MOSAIC	100.00±0.00	100.00±0.00	0.00±0.00
	MOSAIC (2 GT)	100.00±0.00	**100.00±0.00**	0.00±0.00

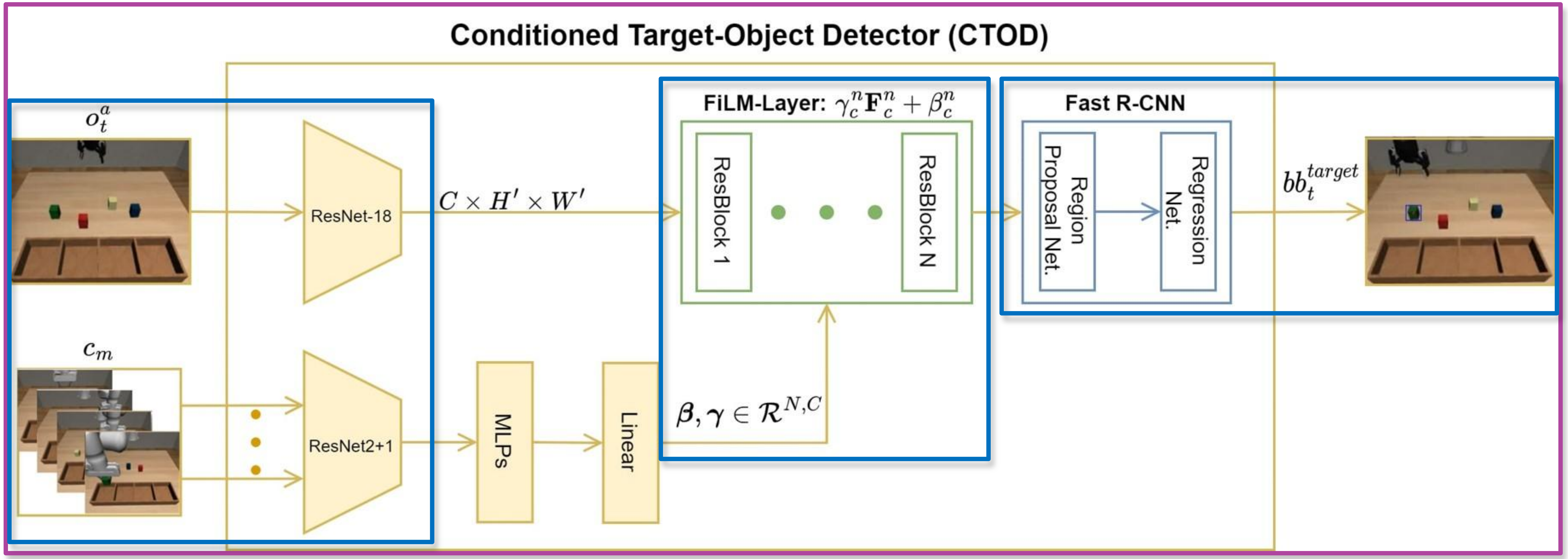
Context:
SoTA Evaluation

- Can we leverage **object prior** to improve system performance?
  - *Dual-stage learning approach*:
    1. **Command Analysis,** $bb_t^{target} = f_\phi(\mathrm{o}_\mathrm{t}^\mathrm{a}, \mathrm{c}_\mathrm{m})$
    2. **Action Generation,** $\pi_\vartheta\ (a_t |\ bb_t^{target}, z_{task})$



Proposal:
Conditioned Target-Object Detector (CTOD)

- CTOD module implements the function f_ϕ



Perez, E., Strub, F., De Vries, H., Dumoulin, V., & Courville, A. (2018, April). Film: Visual reasoning with a general conditioning layer. In Proceedings of the AAAI conference on artificial intelligence (Vol. 32, No. 1).

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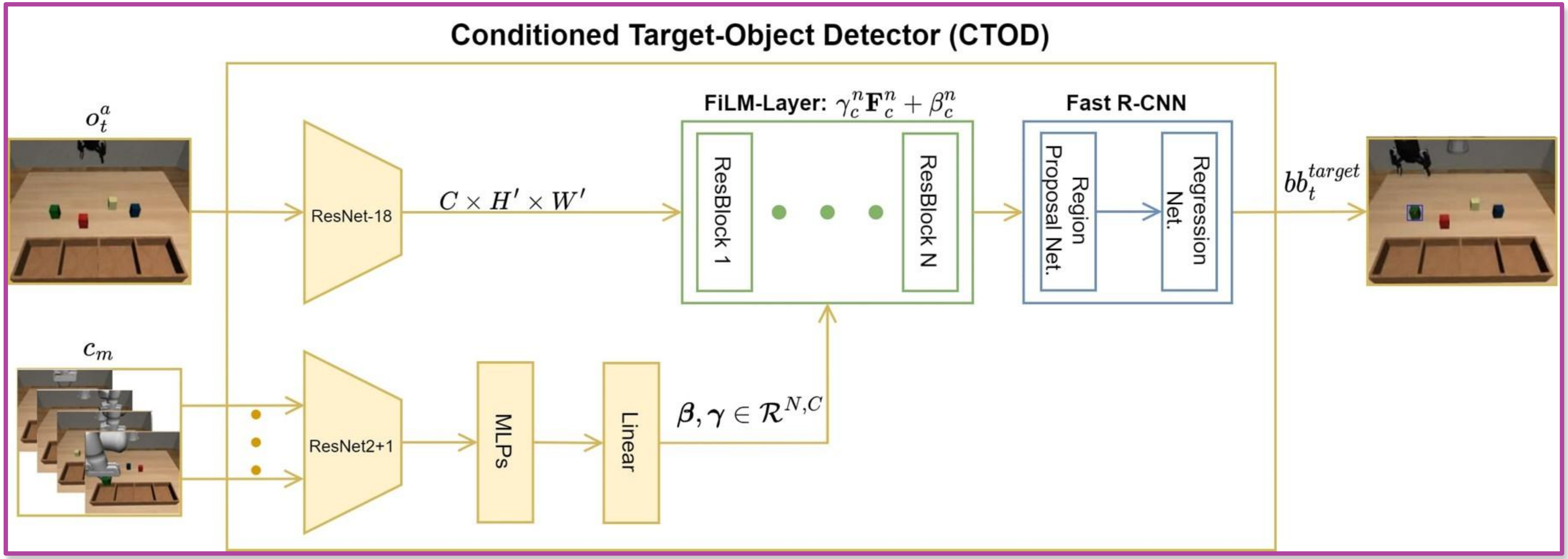
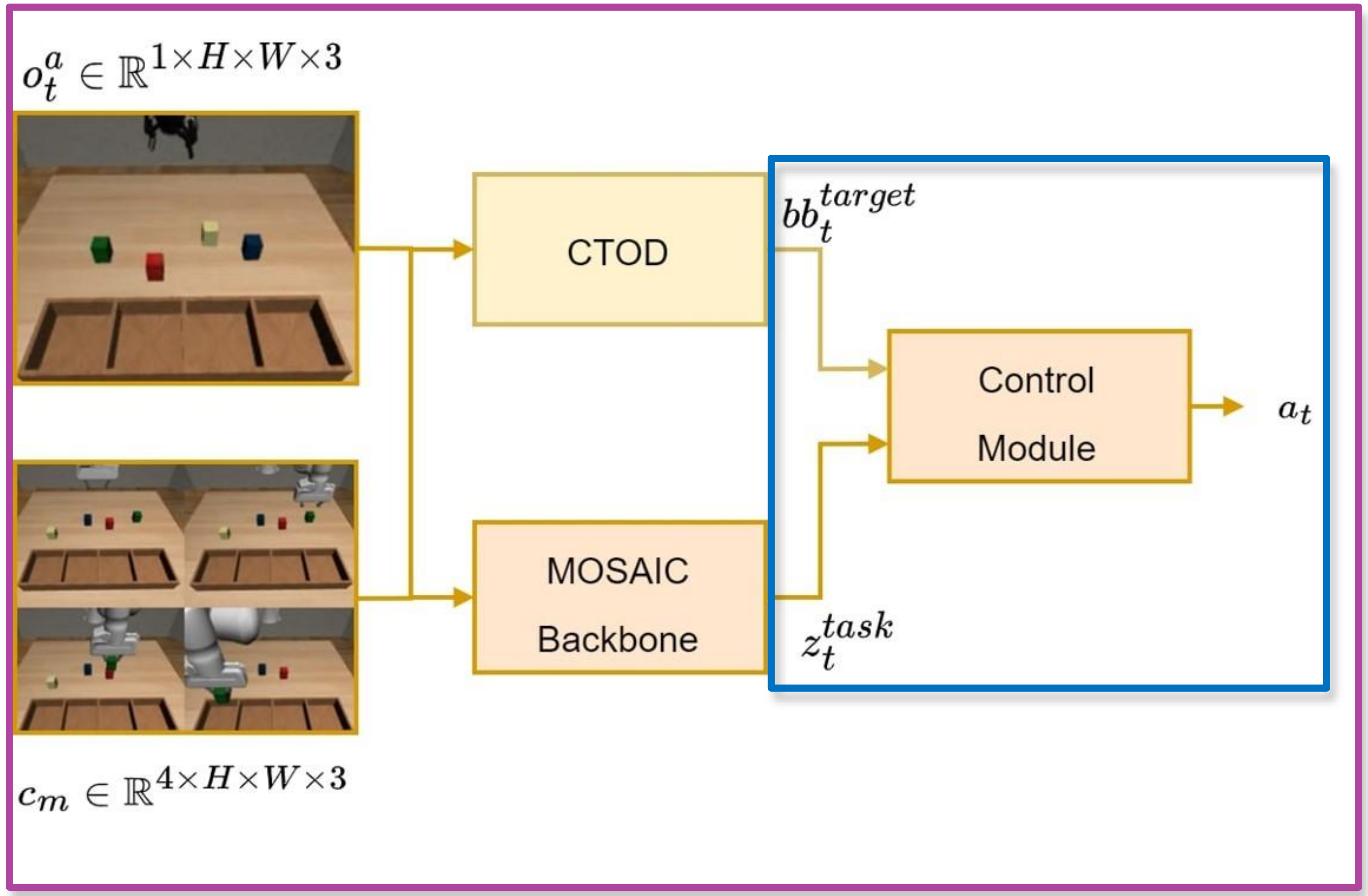


$$\boxed{\mathcal{L}_{CTOD} = \boxed{\gamma_1 \mathcal{L}_{reg}} + \boxed{\gamma_2 \mathcal{L}_{pos}} + \boxed{\gamma_3 \mathcal{L}_{cls}}}$$

Perez, E., Strub, F., De Vries, H., Dumoulin, V., & Courville, A. (2018, April). Film: Visual reasoning with a general conditioning layer. In Proceedings of the AAAI conference on artificial intelligence (Vol. 32, No. 1).

Proposal: MOSAIC-CTOD

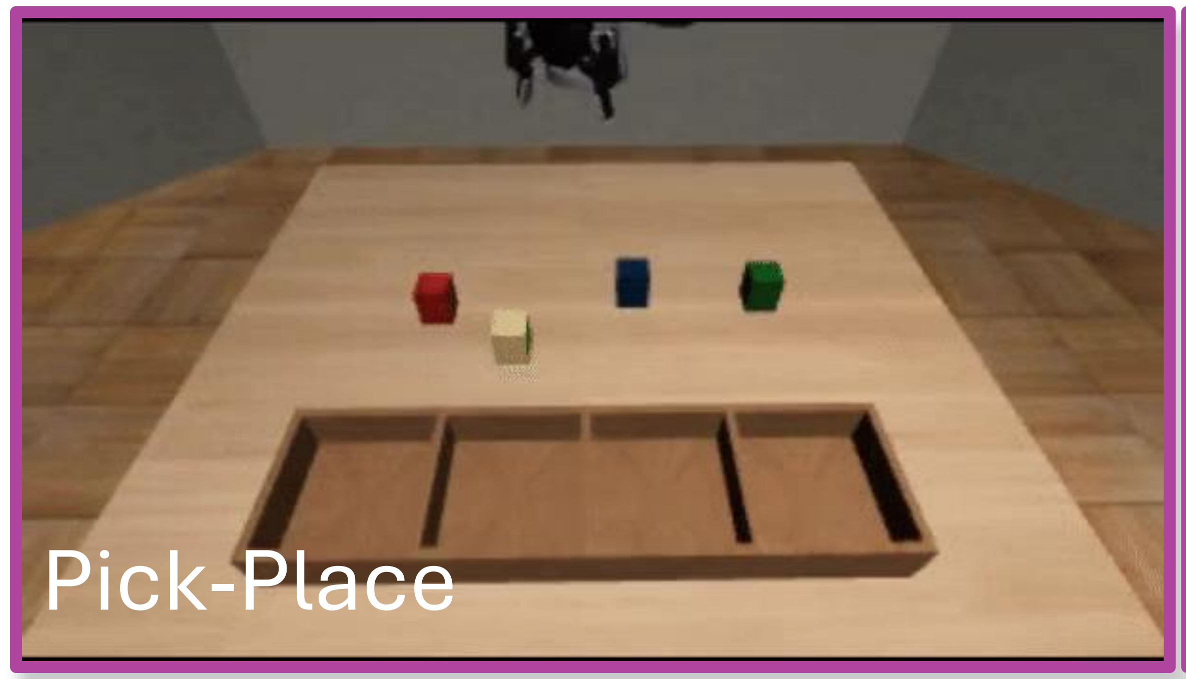
- To model the policy π_ϑ , we enrich the MOSAIC baseline, integrating the **trained** CTOD module
- We focus on **improving** the input of the ***Control Module***, which now receives the predicted target bounding-box and the task embedding
- During learning we freeze the CTOD module and use the same loss function as in MOSAIC
- This adjustment allows the backbone to concentrate on encoding the task intent during the learning process

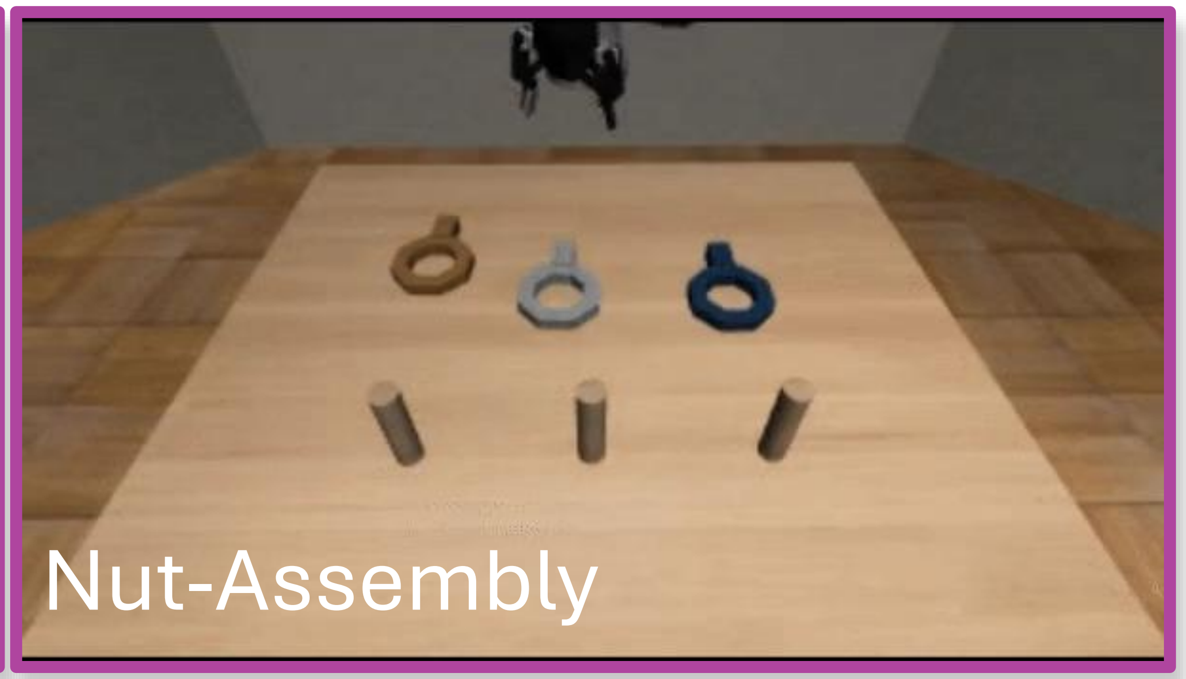


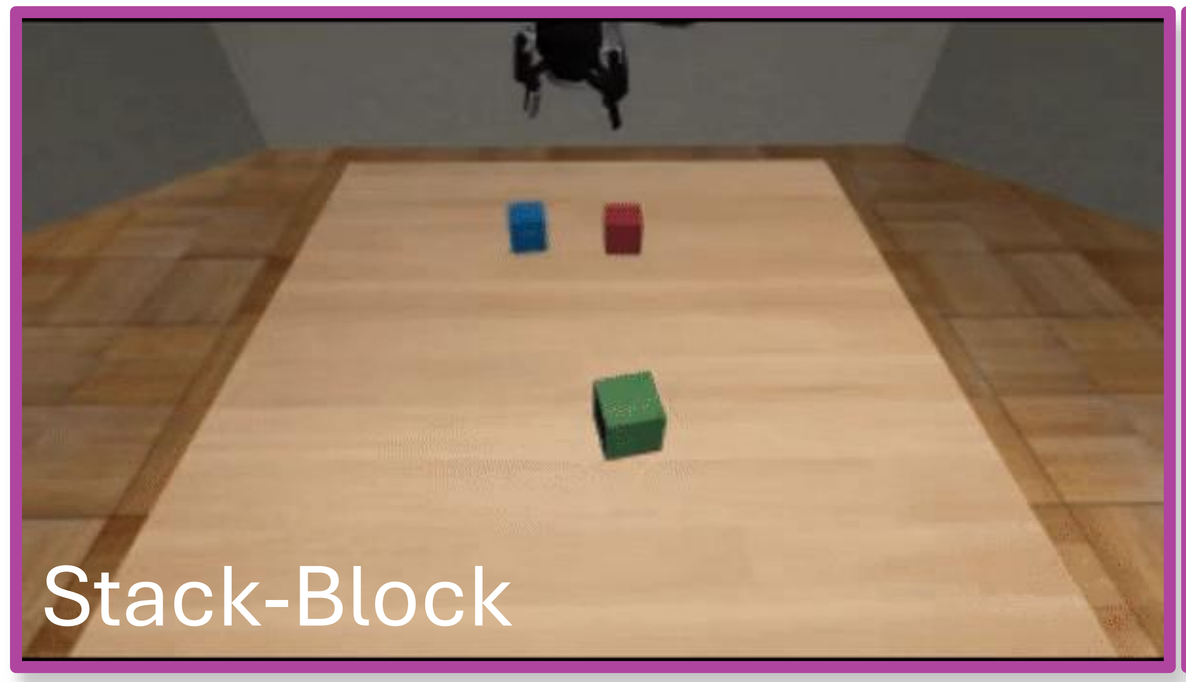
$$\mathcal{L}_{MOSAIC} = \boxed{\gamma_{BC}\mathcal{L}_{BC}} + \boxed{\gamma_{Inv}\mathcal{L}_{Inv}} + \boxed{\gamma_{Rep}\mathcal{L}_{rep}}$$

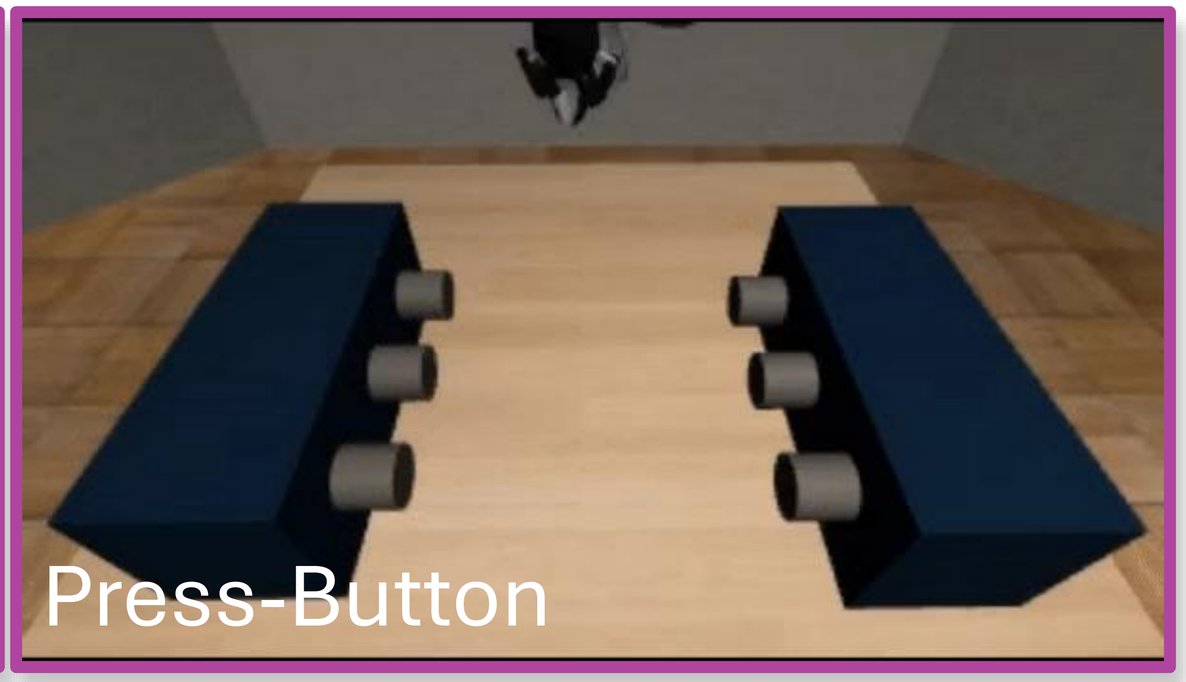
Experimental Results: Dataset

- We consider **4 multi-variation tasks**, for each variation we collect **100** trajectories for the agent and the demonstrator
    - ***Pick-Place***: 16 variations
    - ***Nut-Assembly***: 9 variations
    - ***Stack-Block***: 6 variations
    - ***Press-Button***: 6 variation
- Two test setting:
    - *Single-Task Multi-Variation Setting*
    - *Multi-Task Multi-Variation Setting*









Experimental Results:

Single-Task Multi-Variation Scenario

- For each task we train a specific method
- Improvement on 3 over 4 tasks, w.r.t. MOSAIC baseline:
  - Pick-Place: +18.36%
  - Nut-Assembly: +30.74%
  - Stack-Block:+38.34%
  - Press-Button: -4.44%

Task	Method	Reaching [%]	Success [%]	Wrong Obj. [%]
Pick-Place	TOSIL	33.12±1.08	26.87±0.62	66.04±1.57
	MOSAIC	62.90±0.95	58.75±1.87	37.71±0.72
	MOSAIC (2 GT)	89.17±1.30	84.17±1.57	11.46±1.30
	MOSAIC-GT-BB	100.00±0.00	76.46±3.20	0.00±0.00
	MOSAIC-CTOD	98.12±0.62	**77.11±5.60**	1.04±0.36
Nut-Assembly	TOSIL	36.30±1.28	31.11±1.92	56.30±3.57
	MOSAIC	38.89±1.11	33.33±1.11	51.48±2.31
	MOSAIC (2 GT)	85.19±4.49	78.89±4.01	11.11±2.22
	MOSAIC-GT-BB	100.00±0.00	70.37±1.69	0.00±0.00
	MOSAIC-CTOD	98.89±1.11	**64.07±0.64**	0.00±0.00
Stack-Block	TOSIL	69.44±0.96	48.89±2.54	41.67±1.66
	MOSAIC	60.56±0.96	53.33±0.16	36.11±0.96
	MOSAIC (2 GT)	91.11±0.96	73.89±4.19	7.22±0.96
	MOSAIC-GT-BB	100.00±0.00	90.00±2.88	0.00±0.00
	MOSAIC-CTOD	100.00±0.00	**91.67±2.88**	0.00±0.00
Press-Button	TOSIL	83.33±1.66	75.00±3.33	16.67±1.67
	MOSAIC	100.00±0.00	**100.00±0.00**	0.00±0.00
	MOSAIC (2 GT)	100.00±0.00	100.00±0.00	0.00±0.00
	MOSAIC-GT-BB	92.22±2.54	90.56±1.92	3.88±0.96
	MOSAIC-CTOD	97.22±1.92	95.56±1.92	2.77±0.96

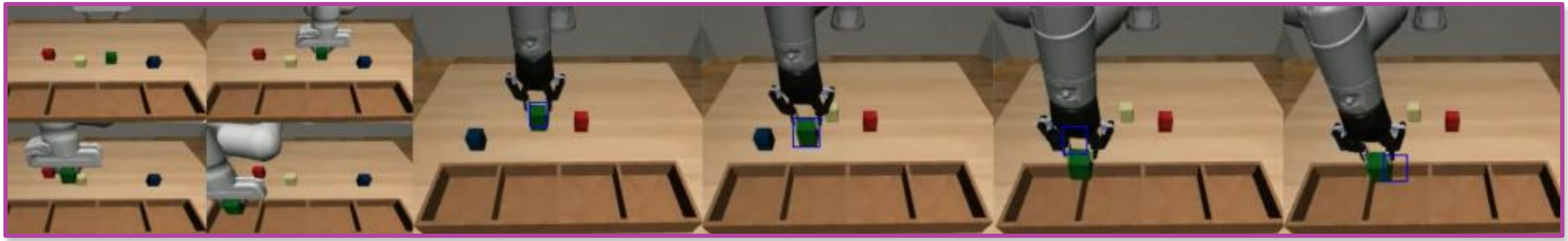
Experimental Results: Multi-Task Multi-Variation Scenario

- We train a single multi-task model
- Improvement on 4 over 4 tasks, w.r.t. MOSAIC baseline:
  - Pick-Place: +30.62%
  - Nut-Assembly: +4.80%
  - Stack-Block:+32.36%
  - Press-Button: +13.89%

Task	Method	Reaching [%]	Success [%]	Wrong Obj. [%]
Pick-Place	TOSIL	35.42±0.72	26.46±1.80	64.17±2.20
	MOSAIC	25.83±1.30	22.71±0.72	67.92±1.30
	MOSAIC (2 GT)	61.04±2.19	56.46±3.14	36.25±1.08
	MOSAIC-GT-BB	100.00±0.00	58.33±0.90	0.00±0.00
	MOSAIC-CTOD	80.21±1.44	**53.33±1.90**	0.37±0.64
Nut-Assembly	TOSIL	27.78±2.94	22.59±3.57	64.81±2.57
	MOSAIC	32.96±1.28	28.53±2.31	41.67±1.66
	MOSAIC (2 GT)	76.67±1.11	65.56±4.00	17.04±1.69
	MOSAIC-GT-BB	99.63±0.64	37.04±5.70	0.00±0.00
	MOSAIC-CTOD	68.68±1.11	**33.33±4.00**	0.00±0.00
Stack-Block	TOSIL	54.44±0.96	48.89±0.96	45.56±0.96
	MOSAIC	57.78±1.92	55.56±2.54	44.81±3.20
	MOSAIC (2 GT)	95.56±0.96	89.44±1.92	4.44±0.96
	MOSAIC-GT-BB	94.44±5.85	73.89±5.00	0.00±0.00
	MOSAIC-CTOD	97.92±2.09	**87.92±2.00**	0.00±0.00
Press-Button	TOSIL	70.56±7.52	60.59±11.34	27.22±2.47
	MOSAIC	78.33±1.66	77.22±2.54	22.78±2.54
	MOSAIC (2 GT)	77.22±0.962	76.67±1.667	23.33±1.66
	MOSAIC-GT-BB	100.00±0.00	97.22±0.96	0.00±0.00
	MOSAIC-CTOD	92.78±0.96	**91.11±2.54**	0.00±0.00

Experimental Results: Error cases and Improvement

1. We have a modular system that can suffer of *error-propagation phenomena*



2. Not uniform learning during multi-task

Single-Task

Task	Method	Success [%]
Pick-Place	MOSAIC-CTOD	77.11±5.60
Nut-Assembly	MOSAIC-CTOD	64.07±0.64
Stack-Block	MOSAIC-CTOD	91.67±2.88
Button	MOSAIC-CTOD	95.56±1.92

Multi-Task

Task	Method	Success [%]
Pick-Place	MOSAIC-CTOD	53.33±1.90
Nut-Assembly	MOSAIC-CTOD	33.33±4.00
Stack-Block	MOSAIC-CTOD	87.92±2.00
Button	MOSAIC-CTOD	91.11±2.54

Conclusion

- This study addresses the challenge of **target object misidentification** within the context of learning from video demonstration
- The proposed CTOD can **identify the target object** by predicting its **class-agnostic bounding box**
- Results demonstrated that the predicted object prior significantly **enhances success rates** in both single-task (average improvement of +20.75%) and multi-task scenarios (average improvement of +20.42%)

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Thanks for the attention

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